MIL-P-48322B (AR)
AMENDMENT 5
21 March 1984
SUPERSEDING
AMENDMENT 4
24 June 1982

MILITARY SPECIFICATION

PROJECTILE, 8 INCH, HE, RA, M650
AFT WARHEAD SECTION AND STRIPPABLE COATING ASSEMBLY

This amendment forms a part of Military Specification MIL-P-48322B (AR), dated 22 May 1981 and is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Preamble: Change the word "Department" to "Departments".

Title: Delete "Section and Strippable Coating".

1.1: Delete "Section and Strippable Coating".

PAGE 2

2.1: Dwg. 9288037, delete, "Section and Strippable Coating".

* Delete: Drawings 9287967 and 9287968 in their entirety.

2.2: Delete "E18 - Test for Rockwell Hardness of Metallic Materials" and substitute the following:

"E10 - Test for the Brinnell Hardness of Metallic Materials".
E103 - Rapid Indentation Hardness Testing of Metallic Materials".

PAGE 3

* 3.4: Delete "(information only)".

3.5.2: Delete "burns" and substitute "burrs".

PAGE 5

* Add the following new paragraph:

"3.15 Knurl. The minimum effective depth of the radial grooves in the knurl shall be .012 inches between the first and sixth peaks (See Figure 2)."

Add "Figure 2" as follows:

FSC 1320

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Reference Dwg. 9287943, "Informational View"

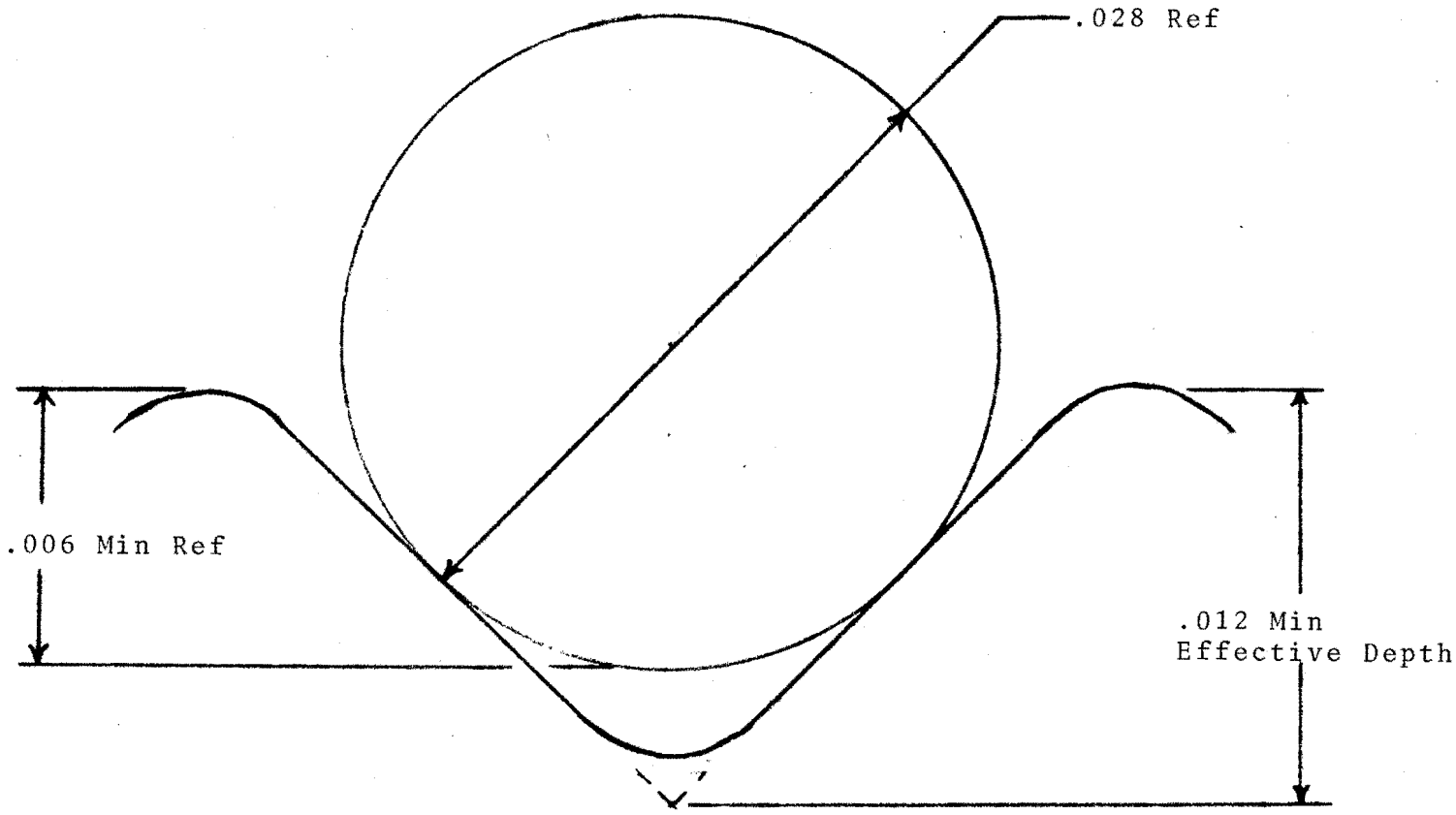


FIGURE 2

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PAGE 6

4.3.1: Delete "Section and Strippable Coating" two places.

Delete: "Disk 9288038 10".

PAGE 7

* Table I: Delete: "Section and Strippable Coating"

Under "Warhead (Prior to Painting) Metal Parts", delete "Interior Surface Finish (c) 3.9 4.5.6".

Delete: "Base Cavity Radius" and substitute "Base Cavity Radius, if as forged".

Add the following: "Precracked Charpy Test 1 3.4 4.5.3/Appendix".

Delete: "Disk (Dwg. 9288038) 4.4.2.8".

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* 4.4.1: Delete "and interior surface and internal radius" and substitute "and internal radius, if as forged".

PAGE 10

4.4.2.2: Delete "Critical defect 1 and Note (a) in their entirety.

* Add Special defect A as follows:

"Special

A Precracked charpy test 4.4.3.2 3.4 4.5.3/Appendix".

* Delete Major defects 103 and 105 in their entirety and substitute the following:

"Major category:

Major 103: Radius of base cavity, if machined 0.40% 3.2 Gage

Major 105: Radius of base cavity, if as forged e 3.2 4.5.5".

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4.4.2.3: Delete Major defect 101 in its entirety and substitute the following:

"Major category:

Major 101: Pitch diameter of base thread, min. 0.40% 3.2 Gage".

Add Minor defects 203 and 204 as follows:

"Minor category:

Minor 203: Pitch diameter of base thread, max. 0.65% 3.2 Gage

Minor 204: Major diameter of base thread, max. 0.65% 3.2 Gage".

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4.4.2.4: Add the following new Major defect:

"Major category:

* Major 113: Delete "Gage" and substitute "Visual/Comparison Blocks."

Add the following Major defects:

"Major category:

Major 117: Width of knurl, min. 0.40% 3.2 Gage

* Major 118: Difference in height of first and eleventh peak of knurl over maximum (measure 3 places, 120° apart) 100% Gage.

* Major 119: Depth of radial grooves in knurl, minimum (check three places), (a) 3.15 4.5.14".

Delete Minor defect 212 in its entirety.

Add the following Minor defects:

"Minor category:

Minor 213: Finish of base insulator cavity 0.65% 3.2 Visual

Minor 214: Surface finish 0.65% 3.9 Visual

Minor 215: Evidence of poor workmanship 1.0% 3.14 Visual

* Minor 216: Minor diameter of nose thread, min. 0.65% 3.2 Gage"

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Add Note (a) as follows:

"(a) See paragraph 4.4.3.11".

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4.4.2.5, Critical defect 2: Change "Visual/4.5.9" to "Visual/Lighting".

PAGE 16

4.4.2.7: Delete Major defect 102 in its entirety and substitute the following:

"Major category:

Major 102: Insulation or adhesive above max. Gage. 0.40% 3.2

* Major defect 103: Delete "100%" and substitute "0.40%".

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Delete paragraph 4.4.2.8 in its entirety.

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4.4.2.9: Delete "Section and Strippable Coating".

Delete Minor defects 201, 202 and 203 in their entirety and substitute the following:

"Minor category:

Minor 201: Corrosion preventive compound missing from threads (both ends) or from knurl 0.65% 3.2 Visual

Minor 202: Protective plug missing from forward end 0.65% 3.2 Visual

Minor 203: Protecitve plug, aft end missing 0.65% 3.2 Visual".

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4.4.2.10: Change Dwg. No. from "9288037" to "9280132".

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4.4.3.1.1: In the third sentence change the word "selected" to "selecting".

* 4.4.3.1.3: In the second sentence change "re-temper" to "reheat treatment".

* Delete 4.4.3.2 in its entirety and substitute the following:

"4.4.3.2 Slow bend precracked charpy test. Beginning with the first heat treat lot produced and continuing until three consecutive heat treat lots have complied with the applicable minimum requirement specified, one final heat treated warhead from each heat treated lot shall be subjected to the precrack charpy test. The warhead selected shall be of the highest average hardness reading from 4.4.3.1.1 and may be the same as used in 4.4.3.1.2. Two specimens shall be tested, with acceptance based on the average crack strength. In addition, no individual crack strength shall be less than 23 KSI in. If one test is invalid, acceptance may be based on the crack strength of the valid sample, or another sample from the same warhead may be tested. If either the average or individual crack strength is below the minimum allowable, the heat treated lot represented shall be rejected, or subjected to reheat treatment and retest, for all mechanical properties, if approved by the Government. After three consecutive heat treated lots have been accepted, the test shall be discontinued. The test shall be re-instituted, until three consecutive heat treated lots have complied with the applicable requirements, if any of the following occur:

a. Steel is procured from a different producer, or from a substantially different process as determined by the Government.

b. Forgings are procured from a different producer.

c. A substantial change is made in the forging process or configuration, as determined by the Government.

d. A change is made in heat treatment contractors.

e. A substantial change is made in the heat treatment process, as determined by the Government."

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4.4.3.3: Delete in its entirety and substitute the following:

"4.4.3.3 Steel quality. Steel supplied shall comply with MIL-STD-50783, except that macroetch quality shall be per MIL-STD-1459. The quality of the steel as indicated by the macroetch results shall be equal to, or better than A5, B3 and C8 with defects D1, D3, D4, D5, D6, D7 or D8 unacceptable. Manufacturer shall provide a certification that the applicable MIL-S-50783, and macroetch requirements have been met."

4.4.3.5: Delete in its entirety.

4.4.3.7: Delete "3.5.11" and substitute "3.11".

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4.4.3.9: Delete the first sentence and substitute the following:

"After qualification in accordance with TTC-490, two salt spray samples shall be taken each week, one during the first 30 minutes of production and a second at any other time."

* Add the following new paragraph:

"4.4.3.11 Depth of knurl

4.4.3.11.1 Initial sampling. The first pieces from each tool, forming the radial groove, shall be inspected for depth until one part has been accepted.

* 4.4.3.11.2 Subsequent sampling. If the life of the tool forming the radial grooves is 50 parts or less, the last part formed by each tool shall be inspected. If tool life is greater than 50 parts, an additional inspection of one part from each 50 shall be performed. If the knurl depth is less than .012 inches, production represented by that sample shall be rejected, production shall be stopped and corrective action taken. Sampling for any tool producing a rejected part shall revert to 4.4.3.11.1 if that tool is returned to production."

* 4.5.1: Delete in its entirety and substitute the following:

"4.5.1 Mechanical properties-hardness. The hardness test shall be conducted in accordance with ASTM Method E 103 and the following:

a. The surface of the body in the area to be hardness

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tested shall be prepared by machining or grinding to a sufficient depth, .015 inches minimum, to remove scale and get below the decarburization zone.

b. Hardness tester periodic verification shall be performd using Brinnell test standards manufactured in accordance with ASTM Method E-10 in lieu of production samples.

c. All hardness test measurements shall be automatically displayed.

d. The hardness tester shall use a standard 10MM Brinnel ball. The body shall be securely supported so that no rocking or shifting of the body will occur during the test. The tester shall apply a preload followed by application of the full load. The preload shall not exceed 10% of the major load of 3000Kg. The amount of penetration of the indentor (10MM ball) during that time period when the major load is applied including the dwell time at the end of the load application shall be automatically measured. The tester shall be provided with a means to compensate for any flexing of the test item that may occur during application of the major load. The hardness number will be displayed at the conclusion of the test.

e. One reading shall be taken at location X and one at location Y as indicated on Dwg. 9287943."

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* 4.5.2: Delete the last sentence, "The contractor....DI-T-1906".

* 4.5.3: Delete in its entirety and substitute the following:

4.5.3 Slow bend precrack charpy. The two specimens shall be taken from locations C1 and C2 on Figure 3. The test shall be conducted at a specimen temperature of -65 degree fahrenheit (^{O}F). The test shall be conducted in accordance with the procedure defined in the Appendix of this specification with the notches machined across the rear face of the specimen. Locations C3 and C4 on Figure 3 shall be used for replacement specimens for locations C1 and C2 respectively."

* 4.5.6: Delete in its entirety.

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* 4.5.7: Delete the first two sentences and substitute the following:

"The unpainted warhead, after nose threading but prior to base threading and knurling, shall be mounted in a suitable test fixture. The warhead will then be subjected to the pressure requirements on the applicable drawing. Any warhead which fails to comply with the applicable requirements shall be classed defective and removed from the lot."

* Add "Figure 3" as follows:

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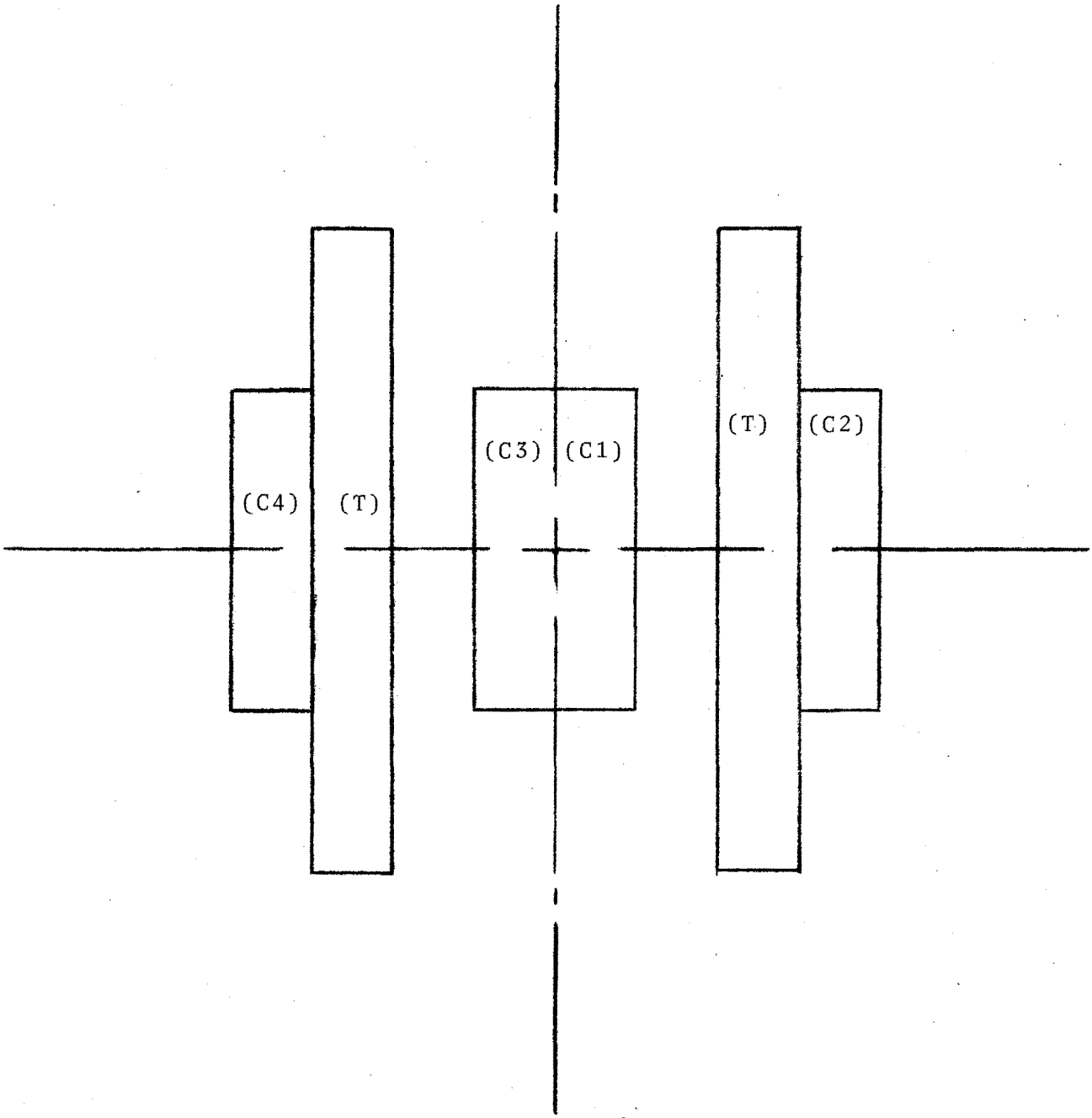


FIGURE 3
Approximate Location of Tension and Charpy Base Specimens
(T- Tensile, C - Charpy)

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4.5.8: Delete the first two sentences and substitute the following:

"The warheads shall be ultrasonically inspected after threading the nose but prior to threading and knurling the base using equipment and procedures in accordance with 4.5.8.1. With the exception of the forward threaded area, the warhead shall be inspected with ultrasonics throughout all of the material."

* 4.5.8.2: Delete "Calibration" and substitute "Verification".

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* 4.5.8.2: Add the following at the end of the paragraph:

"The verification check may be conducted by the equipment operator. Any adjustment required shall be performed by a Level II certified individual (See 4.5.8.4)."

4.5.8.4: Delete in its entirety and substitute the following:

* "4.5.8.4 Operator qualifications. For operation of ultrasonic equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level I. For set-up and calibration of ultrasonic equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level II."

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* Add new paragraph 4.5.14 as follows:

"4.5.14 Depth of knurl. The depth of the radial groove in the knurl shall be checked using a dial indicating depth gage or approved equivalent."

5.1: Change "Level C" to "Level B".

Delete: "Section and Strippable Coating".

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6.5: Change "measured to gaged" to "measured or gaged".

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6.14: Delete in its entirety.

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